INTERNATIONAL MONETARY FUND

NIGER

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THIRD REVIEW UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT, REQUEST FOR WAIVER OF NONOBSERVANCE OF PERFORMANCE CRITERION, AND REQUEST FOR AUGMENTATION OF ACCESS—DEBT SUSTAINABILITY ANALYSIS

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Niger: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate*
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Moderate risk tool: Limited space to absorb shocks*
Application of judgement	*No*

Niger's risk of external and overall public debt distress is rated "moderate" as in the previous DSA. While all thresholds are observed in the baseline, the PV of external debt-to-exports ratio still breaches its threshold under stress test scenarios. Debt-carrying capacity continues to be rated "medium" according to the new methodology. The analysis shows that Niger has limited space to accommodate negative shocks. It remains particularly vulnerable to adverse developments of its exports, which are dominated by uranium and oil products. The debt sustainability assessment is predicated on the government continuing to implement its reform program: fiscal consolidation; structural reforms, including measures to boost revenue mobilization, to contain expenditures, and to diversify the economy; and timely completion of several large-scale investment projects. Identified weaknesses call for further strengthening of debt management, prioritizing concessional borrowing, strengthening private-sector development, and expanding the export base to mitigate the risks associated with commodity price fluctuations.

PUBLIC DEBT COVERAGE

1. The coverage of the public sector is in line with the previous DSA. It includes the central, state and local governments, the social security fund and extra budgetary funds, the central bank, and state guarantees extended to the public and private sectors (Text Table 1). The state and local governments are not holding debt although they are able to directly borrow. The authorities also confirmed the absence of extra budgetary funds and external borrowing by the social security fund (CNSS). Private external debt guaranteed by the government is limited to the guarantee issued to CNPC (China) for the construction of the refinery SORAZ.[1] Government-guaranteed SOE debt mainly consists of loans contracted by the electricity (NIGELEC), water (SPEN) and telecom (Niger Telecom) companies with Exim Bank China, as well as borrowing from the World Bank by ABK, a public administrative entity set up for implementing the Kandadji dam project. External debt is defined on a currency basis.

Text Table 1. Niger: Coverage of Public-Sector Debt and Design of Contingent Liability Stress Test

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The general government, central bank, government-guaranteed debt			
		Default		Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
4	PPP	35	percent of PPP stock	0.0	New PPP law closely scrutinizing any new PPP arrangement.
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

2. The contingent liability tailored stress tests exclusively reflect uncertainty about the vulnerabilities associated with non-guaranteed SOE debt and financial market risks (Text Table 1). For the latter, the shock is kept at the minimum default value of 5 percent of GDP given the small size and depth of the financial sector in Niger. The contingent liabilities shock from SOE debt is set at the default value of 2 percent to reflect risks associated with non-guaranteed SOE debt, currently excluded from the baseline analysis due to data constraints. Coverage will be expanded as the Ministry of Finance collects and processes SOE financial statements with the expected launch of a survey of financial indicators in 2019. Finally, following the adoption of the new PPP law in May 2018, the degree of exposure to PPP-related risks going forward is expected to be well contained. The legislation is in line with good international practices, fully integrates PPPs into the budget cycle, and requires any additional PPP to be subjected to parliamentary approval after being closely scrutinized for value-for-money, and consistency with sectoral

[1] CNPC extended a USD 880 million (CFAF 486.7 billion, 9.3 percent of GDP) loan for the construction of SORAZ refinery in 2008, of which USD 352 million (CFAF 194.7 billion, 3.7 percent of GDP) is guaranteed by the government. The outstanding stock of CFAF 28.3 billion at end-June 2018 is scheduled to be fully repaid by 2019. The baseline stock of debt includes SORAZ debt.

strategies and financial affordability. The debt stock of PPP arrangements contracted under the previous regime amounts to CFAF 214.6 billion (4.5 percent of GDP) at end-December 2017 and is expected to be fully repaid by 2033.

BACKGROUND

A. Evolution and Composition of Debt

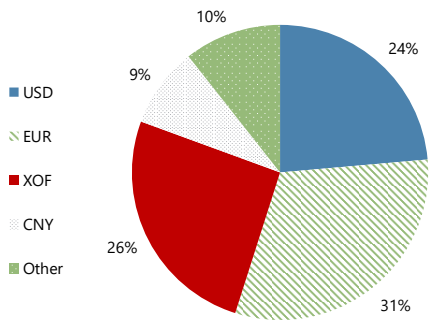
3. Niger's PPG debt stood at CFAF 2,115 billion in June 2018, with external debt accounting for 72 percent of the total according to the latest government publication (Text Table 2). Multilateral creditors represent the lion's share of total public debt, with Niger borrowing most from the World Bank (IDA). Bilateral debt is dominated by China and France (AFD). Around three-quarters of external debt is foreign-currency denominated with relatively low exposure to exchange rate risk given the CFAF's peg to the euro (Text Figure 1). The average weighted interest rate came to 1.6 percent for external debt due to the preponderance of concessional borrowing, and 6 percent for domestic debt. The latter consists mostly of government issued securities, with the role of arrears and statutory advances from the regional central bank declining over time as Niger clears its remaining domestic payments arrears in 2018 and gradually repays loans from the BCEAO. However, debt reported in the government publication does not cover debt related to commercial PPPs and the latest update of IDA debt.

Text Table 2. Niger: Public and Publicly-guaranteed Debt, 2016-June 2018

	Dec-16		Dec-17		Jun-18	
	CFAF bln	percent of GDP	CFAF bln	percent of GDP	CFAF bln	percent of GDP
A. Public debt contracted and disbursed (B+C)	**1793**	**39.8**	**1986**	**41.0**	**2087**	**40.6**
B. Public External debt	**1292**	**28.7**	**1405**	**28.7**	**1493**	**29.1**
Multilateral	1051	23.3	1177	24.9	1232	24.0
Bilateral	242	5.4	228	4.8	262	5.1
Paris Club	46	1.0	54	1.1	59	1.1
Non-Paris Club	196	4.3	174	3.7	203	3.9
C. Public Domestic debt	**501**	**11.1**	**581**	**12.3**	**593**	**11.6**
Tbills and Bonds	422	9.4	519	11.0	541	10.5
Domestic arrears	44	1.0	37	0.8	35	0.7
Other	34	0.8	26	0.5	17	0.3
D. Publicly-guaranteed external debt (SORAZ)	**78**	**1.8**	**41**	**0.9**	**28**	**0.5**
Total PPG debt (A+ D)	**1871**	**41.5**	**2027**	**41.9**	**2115**	**41.2**

Source: IMF staff calculations based on Nigerien authorities' June 2018 debt report.

Text Figure 1. Niger: External Debt by Currency
(share of total external debt as of June 2018)



Sources: Nigerien authorities; and IMF staff calculations.

4. The estimation and analysis of private external debt is complicated by data issues and requires further follow up. The BCEAO does not yet compile external debt stock statistics. Efforts to gather information on the coverage and composition of private external debt will continue, with technical support from the IMF's Statistics Department.

B. Macroeconomic Forecast

5. The baseline scenario is predicated on macroeconomic assumptions reflecting recent economic developments and expected effects of ongoing and new policy measures (Text Table 3). The framework assumes fiscal consolidation in line with the government's ongoing reform program and the goal to meet the WAEMU deficit target of 3 percent of GDP by 2020. Compared to the previous DSA, the commencement of crude-oil exports and the resumption of activities at the Imouraren uranium site have been pushed back by two and four years to 2022 and 2025, respectively, to account for delays in the start of the oil pipeline construction and unfavorable uranium prices. Medium and long-run real GDP growth is expected to be higher relative to the previous DSA on the back of a booming construction sector driven by large investment projects, and the completion of the Kandadji dam and MCC-related activities expected to enhance agricultural productivity. Revenue projections have been dialed back somewhat, but revenue mobilization is expected to gather steam with the implementation of various tax measures included in the past and current budgets, the collection of tax arrears, the consolidation and streamlining of tax exemptions, the fight against petroleum smuggling, and a comprehensive program to overhaul customs and tax administrations. On average, inflation is expected to be somewhat higher over the medium term owing to transitory shocks in 2018, but longer-term projections remain largely unchanged from the previous DSA.

Text Table 3. Niger: Key Macroeconomic Assumptions, 2016-38

	2016-2017	2018-2023	2024-2038 /1
Real GDP growth (percent)			
DSA 2018	4.9	6.9	6.0
Previous DSA	5.0	6.2	5.1
Inflation (CPI)			
DSA 2018	1.3	1.7	2.0
Previous DSA	1.5	2.0	2.0
Primary fiscal balance (percent of GDP)			
DSA 2018	-4.9	-2.0	-1.3
Previous DSA	-6.0	-1.4	0.4
Total revenue excluding grants (percent of GDP)			
DSA 2018	14.3	17.4	19.7
Previous DSA	15.7	19.2	21.9
Exports of goods and services (percent of GDP)			
DSA 2018	16.7	19.0	25.6
Previous DSA	17.4	21.4	23.9
Oil export price (US dollars per barrel)			
DSA 2018	45.4	59.5	60.2
Previous DSA	44.6	42.7	54.3
Uranium price (Thousands of CFAF per kg)			
DSA 2018	54.2	37.5	71.2
Previous DSA	58.2	61.2	75.3

Source: IMF staff calculations.
1/ Averages for the previous DSA are up to 2036.

6. Financing assumptions imply a gradual shift from domestic funding sources to foreign ones. In the short and medium term, Niger is expected to benefit from continued foreign support in line with the pledges made during the December 2017 donor roundtable and investors' forum in Paris.[2] However, in the very long run as economic and structural reforms bear fruit and the country develops, domestic financing should start playing a somewhat larger role again.

7. The composition of foreign and domestic borrowing is also assumed to change over time. For foreign debt, new disbursements are expected to be covered by external funding sources based on historical financing patterns. Over the longer term, the weights of external creditors are adjusted so that external borrowing moves toward less concessional financing and toward commercial loans. For domestic borrowing, debt instruments are assumed to gradually shift from T-bills to medium- and long-term bonds. Consistent with the borrowing terms of recent government securities issuances on the regional market, the average interest rate on domestic debt is assumed at 6.25 and 6.5 percent for bonds maturing in 1 to 3 and 4 to 7 years respectively. The interest rate on T-bills is set to 6 percent.

[2] Niger obtained more funding than sought for its five-year economic and social development plan (PDES 2017-2021), with a total of USD 12.7 billion pledged by official bilateral and multilateral donors while private investors expressed a commitment of USD 10.3 billion.

8. Realism tools flag some deviations from historical experience which reflect mainly a strengthening of Niger's macroeconomic performance.

a. **Drivers of debt dynamics (Figure 3).** Differences between the current and previous DSA reflect updated debt stock data through end-2017, with Niger having taken on less external public debt in the last few years than previously expected. The contributions of past and projected debt-creating flows remain broadly unchanged, although prices and exchange rates are expected to negatively contribute to external debt accumulation relative to historical experience. Relatedly, the projected contribution of real GDP growth to public debt reduction is higher than what the past five years would suggest, due to an upward revision of medium- and long-run growth. Continued fiscal consolidation should curb the contribution of the primary deficit to public debt accumulation more than in the past. For PPG external debt, past forecast errors were mainly driven by unexpected current account, FDI, and price changes, as well as the residual. For public debt, unexpected changes in the primary deficit and the residual mostly accounted for past forecast errors. A comparison with the distribution of past forecast errors for LICs shows that unexpected changes in debt for Niger are within the interquartile range for both public and PPG external debt. .

b. **Realism of planned fiscal adjustment (Figure 4).** The projected 3-year fiscal adjustment in the primary balance (3 percentage points of GDP) lies at the lower end of the top quartile of the distribution of past adjustments of the primary fiscal deficit (above 2.5 percentage points of GDP) derived from the sample of LICs, hence alleviating any concerns of credibility of baseline assumptions. The realism of the expected adjustment is predicated on the authorities' commitment to boost revenue mobilization through the implementation of existing and new measures, to clear domestic payments arrears, and to contain expenditures.

c. **Consistency between fiscal adjustment and growth (Figure 4).** The realism tool flags potentially overoptimistic growth projections considering pronounced fiscal consolidation built into the baseline. However, real GDP growth is expected to reach 5.2 and 6.5 in 2018 and 2019, respectively, in the baseline scenario, on the back of strong activity in the construction and services sectors in the runup to the AU summit and the commencement of large donor-financed investment projects, notably the Kandadji dam and the export pipeline for crude oil.

d. **Consistency between public investment and growth (Figure 4).** The tool shows a similar contribution of public investment to growth across the previous and the current DSAs. Public investment is expected to remain at around 15 percent of GDP in the medium term.

C. Country Classification and Determination of Stress Test Scenarios

9. Niger's debt-carrying capacity remains rated "medium" according to the October 2018 vintage of the World Economic Outlook (WEO). The debt-carrying capacity classification was solely informed by a CPIA rating of 3.41 in the previous framework. The new methodology relies instead on a composite indicator (CI) combining the CPIA score, external conditions as captured by world economic growth and country-specific factors. Based on data from the October 2018 WEO vintage, it yields a CI value of 2.92, reflecting positive contributions from the CPIA (45 percent) but also international reserves (34 percent), and country and world real growth rates (5 and 17 percent respectively) (Text Table 4). This score falls within the medium debt-carrying capacity thresholds defined as 2.69 < CI ≤ 3.05. Debt burden thresholds implied by the medium debt-carrying capacity under the previous and new frameworks are summarized in Text Table 5.

Text Table 4. Niger: CI Score

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.388	1.30	45%
Real growth rate (in percent)	2.719	5.625	0.15	5%
Import coverage of reserves (in percent)	4.052	40.208	1.63	56%
Import coverage of reserves^2 (in percent)	-3.990	16.167	-0.65	-22%
Remittances (in percent)	2.022	0.000	0.00	0%
World economic growth (in percent)	13.520	3.579	0.48	17%
CI Score			2.92	100%
CI rating			Medium	

Source: IMF staff calculations. The CI cutoff for medium debt-carrying capacity is 2.69 < CI ≤ 3.05.

10. The debt sustainability analysis relies on the six standardized stress tests and a tailored commodity price shock stress test. The standardized stress tests use the default settings. While Niger does not qualify for the market financing shock stress test, the commodity price shock stress test is relevant and helps assess the sensitivity of projected debt burden indicators to a sudden one standard deviation decline in commodity export prices.[3]

[3] Niger's moderate risk rating of external and overall public debt distress remains robust to larger customized price shocks including up to two standard deviations.

Text Table 5. Niger: Medium-Debt Carrying Capacity and Debt Thresholds

	PV of PPG external debt in percent of		PPG external debt service in percent of		PV of total public debt in percent of
	GDP	Exports	Exports	Revenue	GDP
Previous framework	40	150	20	20	56
New framework	40	180	15	18	55

Source: IMF staff calculations.

DEBT SUSTAINABILITY

A. External Debt Sustainability

11. External debt is projected to fall gradually, with public and private debt both declining in the long run (Table 1). Under the baseline scenario, PPG external debt is expected to increase from 34 to 36.6 percent of GDP over 2018-21 owing to significant foreign borrowing to finance Niger's economic and social development agenda, before falling to 28.5 percent of GDP by 2038. The non-interest current account deficit remains the main driver of external debt dynamics, with the goods and services balance likely deteriorating in the short term due to high imports related to the construction boom ahead of the 2019 AU summit. In the medium term, large investment projects, including the Kandadji dam, a cement factory and the oil export pipeline, and MCC-funded investments in the agricultural sector should entail higher current account deficits. Once the non-interest current account deficit, net FDI and endogenous debt dynamics are accounted for, remaining drivers of external debt accumulation/reduction such as components of the capital account, reserves accumulation, valuation adjustments as well as price and exchange rate changes are subsumed into the residual.

12. None of Niger's PPG external debt indicators breaches thresholds under the baseline scenario (Figure 1 and Table 3). Similar to the previous DSA results, all indicators remain below applicable debt thresholds throughout the projection period. The PV of debt-to-GDP ratio is expected to hover around 24 percent of GDP in the longer term. After peaking in 2021, the PV of debt-to-exports ratio is projected to decline with the commencement of crude-oil exports following the completion of the export pipeline, and the expected resumption of the Imouraren uranium project after 2025. Both the debt service-to-exports and debt service-to-revenue ratios display similar patterns, declining sharply in 2020 with the full amortization of debt owed to CNPC, picking up in 2021 owing to principal repayment on IMF and IDA loans, and decreasing gradually thereafter. For these indicators, the baseline scenario yields a debt path very close to that in the historical scenario. But longer-term projections of the PV of debt-to-GDP and PV of debt-to-exports ratios point to a rapidly accumulating debt if key macroeconomic variables were kept at their historical averages.

13. However, the PV of external debt-to-exports ratio breaches its applicable threshold under the export shock stress test scenario, thereby signaling a moderate risk of external debt distress. The

indicator peaks at 253 percent in 2021 before progressively receding below its prescribed threshold, helped by the anticipated large boost to oil exports. This outcome is consistent across DSA vintages and underscores Niger's narrow export base concentrated on uranium and oil, generating structural vulnerability to unfavorable commodity prices and unstable export revenues. The threshold is also breached under the stress test combining adverse shocks to the current transfers-to-GDP and FDI-to-GDP ratios, albeit only in 2020 and 2021; the stress test combining the five standardized stress tests in 2021; and the tailored commodity price shock stress test over 2019-21.

14. Alternative scenarios incorporating delays in the commencement of crude oil exports or the resumption of activities at the Imouraren uranium site do not change Niger's external and overall debt ratings. If uranium operations resume in 2025 as in the baseline but crude-oil exports only start in 2023 instead of 2022, the PV of external debt-to-exports ratio would breach its prescribed threshold through only an extra year to 2023 under the export shock stress test scenario and would subsequently decline as in the baseline case without the delay. While the breach is prolonged through 2033 in the extreme scenario where the pipeline construction is not completed over the DSA horizon and the expected boost to exports does not materialize, the PV of external debt-to-exports ratio remains below its debt threshold applicable under the baseline scenario so that the rating of external debt distress would remain "moderate". Assuming that crude oil exports commence in 2022, a similar exercise with uranium exports delayed by one year to 2026 shows that the threshold is respected under the export shock stress test from 2022 onwards. If the recovery of the uranium sector is postponed to after 2038, the PV of external debt-to-exports ratio remains below its threshold applicable under the baseline scenario. Also, the breach of the threshold under the export shock stress test scenario is ended in 2022 as in the baseline, though the indicator stays close to its threshold.

B. Domestic Debt Sustainability

15. Public sector debt is projected to build up somewhat in the medium term, driven by foreign borrowing before declining over the longer term (Table 2). While public debt is expected to increase with the spending of donor funds, it should recede with the government's commitment to fiscal consolidation and meeting the WAEMU fiscal convergence criterion by 2020. Accordingly, the primary deficit is expected to decline over time with the improvement of domestic revenue mobilization, enhancement of spending quality, and continued expenditure control, further helped by favorable automatic debt dynamics.

16. The PV of public debt-to-GDP ratio breaches its benchmark neither under the baseline, nor the stress test scenarios (Figure 2 and Table 4). Similar to the previous DSA, public debt indicators in the baseline scenario remain below their levels suggested in the historical scenario. Again, the commodity price shock emerges as the most extreme stress test among all alternative scenarios, emphasizing the vulnerability of the public debt path to commodity price fluctuations.

C. Risk Rating and Vulnerabilities

17. Niger's debt sustainability analysis finds a moderate risk of external and overall debt distress. The baseline scenario indicates a sustainable debt path for both PPG external debt and public

debt, but the prescribed thresholds are breached in the event of a negative shock to export growth or commodity prices.

18. A granular assessment of the moderate risk rating shows that Niger has limited space to absorb shocks (Figure 5). While the debt service-to-revenue ratio is well below its threshold, such that only shocks in the upper quartile of the observed distribution of shocks, i.e. larger than 35 percent of the threshold, would push the ratio above its prescribed limit, the occurrence of the median shock suffices to downgrade Niger's risk rating to "high" by pushing the PV of debt-to-exports ratio above its threshold in 2020 and 2021. The remaining indicators are sensitive to shocks larger than the median, albeit to different degrees. Such shocks would jeopardize the debt path by pushing the PV of debt-to-GDP ratio above its threshold from 2020 onward, while the debt service-to-exports ratio would breach its threshold in 2018 and 2019 only.

19. Debt sustainability is contingent on the reforms built into the baseline being implemented. Commitment to fiscal consolidation and steadfast implementation of the structural reforms are instrumental to contain debt accumulation and preserve macroeconomic stability. The baseline also presumes that Niger's large investment projects are successfully completed on time. Economic diversification supported by private-sector development and financial deepening is crucial for expanding the export base and mitigating the risks associated with commodity price fluctuations. Identified weaknesses call for further strengthening of debt management and sticking to a prudent borrowing strategy that prioritizes concessional borrowing.

D. Authorities' Views

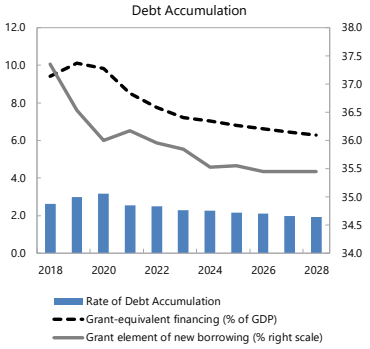
20. The authorities broadly agreed with the assumptions and results of the DSA and made a few observations. While noting that the envisaged commencement of crude-oil exports would further increase Niger's reliance on oil, they acknowledged the importance of diversifying the export base, mostly at the intensive margin, by raising the value-added content of existing exports. Developing manufacturing industries and modernizing existing value chains would allow exporting transformed products instead of low value-added primary commodities. They highlighted the mitigating role of Niger's access to WAEMU's pooled reserves. They also noted that the default size of 2 percent of GDP used in the contingent liability stress test tailored to capture SOE debt appears rather large given their assessment of a negligible risk associated with SOE borrowing from the domestic banking system, the limited number of SOEs that are able to borrow abroad (due to their small size and poor capacity to produce reliable financial statements), and the inclusion of government-guaranteed external debt of major SOEs in the analysis. They expressed their commitment to extend public sector coverage by providing the end-December 2017 debt stocks of major SOEs vis-à-vis the banking sector for the next review.

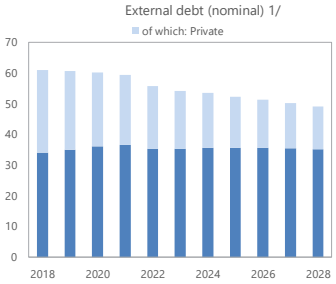
Table 1. Niger: External Debt Sustainability Framework, Baseline Scenario, 2015-2038

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	**61.4**	**64.4**	**62.5**	**61.0**	**60.6**	**60.2**	**59.3**	**55.8**	**54.2**	**49.2**	**36.6**	**50.2**	**55.2**
of which: public and publicly guaranteed (PPG)	**30.1**	**32.6**	**32.4**	**34.0**	**34.9**	**36.1**	**36.6**	**35.4**	**35.2**	**35.2**	**28.5**	**23.5**	**35.4**
Change in external debt	10.3	3.0	-1.9	-1.6	-0.4	-0.4	-0.8	-3.6	-1.6	-1.1	-1.4		
Identified net debt-creating flows	**20.5**	**9.2**	**8.3**	**11.1**	**10.5**	**10.9**	**9.6**	**2.9**	**4.8**	**3.3**	**2.8**	**6.7**	**6.3**
Non-interest current account deficit	**20.4**	**15.5**	**16.2**	**18.7**	**20.6**	**22.8**	**18.9**	**14.1**	**13.3**	**11.0**	**9.9**	**18.5**	**15.2**
Deficit in balance of goods and services	22.4	17.0	18.7	21.0	23.3	25.4	21.3	14.8	12.8	10.1	8.3	**21.1**	**15.8**
Exports	18.3	16.2	17.2	16.7	17.0	17.0	17.1	22.5	23.7	25.8	24.9		
Imports	40.7	33.2	35.9	37.8	40.3	42.5	38.4	37.4	36.5	35.9	33.2		
Net current transfers (negative = inflow)	-4.1	-3.7	-4.7	-4.0	-4.3	-4.3	-4.0	-3.6	-3.5	-3.1	-2.5	**-4.0**	**-3.6**
of which: official	-2.2	-1.8	-2.9	-2.0	-2.5	-2.4	-2.1	-1.8	-1.8	-1.8	-1.6		
Other current account flows (negative = net inflow)	2.1	2.1	2.1	1.7	1.7	1.7	1.6	2.8	4.0	4.0	4.1	**1.3**	**3.0**
Net FDI (negative = inflow)	**-6.8**	**-3.5**	**-3.7**	**-5.3**	**-6.9**	**-9.1**	**-6.7**	**-5.9**	**-5.5**	**-5.4**	**-5.4**	**-9.6**	**-6.0**
Endogenous debt dynamics 2/	**7.0**	**-2.9**	**-4.2**	**-2.4**	**-3.2**	**-2.8**	**-2.6**	**-5.3**	**-3.0**	**-2.2**	**-1.7**		
Contribution from nominal interest rate	0.0	0.0	0.0	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5		
Contribution from real GDP growth	-2.5	-2.9	-2.9	-2.8	-3.7	-3.3	-3.1	-5.8	-3.5	-2.8	-2.1		
Contribution from price and exchange rate changes	9.5	0.0	-1.2	...	...	...	...	...	...	...	...		
Residual 3/	**-10.2**	**-6.2**	**-10.2**	**-12.6**	**-10.9**	**-11.4**	**-10.5**	**-6.5**	**-6.4**	**-4.4**	**-4.2**	**-2.3**	**-7.5**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**23.1**	**22.5**	**23.7**	**24.6**	**25.0**	**24.2**	**24.2**	**24.7**	**20.6**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**134.4**	**134.7**	**139.8**	**144.1**	**146.1**	**107.5**	**102.3**	**95.9**	**82.7**		
PPG debt service-to-exports ratio	**3.2**	**4.0**	**4.3**	**10.1**	**10.5**	**8.2**	**9.4**	**7.0**	**6.6**	**6.3**	**6.3**		
PPG debt service-to-revenue ratio	**3.3**	**4.5**	**5.2**	**10.3**	**11.2**	**8.3**	**9.1**	**8.5**	**8.2**	**8.4**	**7.7**		
Gross external financing need (Million of U.S. dollars)	1028.6	967.5	1073.5	1407.5	1549.8	1659.6	1637.8	1311.6	1380.3	1550.3	2911.1		
Key macroeconomic assumptions													
Real GDP growth (in percent)	4.3	4.9	4.9	5.2	6.5	6.0	5.6	11.0	6.8	6.0	6.1	**5.8**	**6.4**
GDP deflator in US dollar terms (change in percent)	-15.6	0.0	1.9	8.5	1.0	3.3	2.6	2.3	2.4	2.0	2.4	**1.1**	**2.7**
Effective interest rate (percent) 4/	0.0	0.0	0.0	0.8	0.8	0.9	0.9	0.9	1.0	1.2	1.3	**0.8**	**1.0**
Growth of exports of G&S (US dollar terms, in percent)	-23.5	-7.1	13.6	11.1	9.2	9.9	8.8	49.4	15.1	9.8	7.6	**7.5**	**13.9**
Growth of imports of G&S (US dollar terms, in percent)	-7.9	-14.4	15.6	20.0	14.7	15.5	-2.1	10.5	6.9	7.5	7.4	**10.0**	**9.4**
Grant element of new public sector borrowing (in percent)	...	...	...	37.4	36.5	36.0	36.2	36.0	35.8	35.4	35.4	**...**	**35.9**
Government revenues (excluding grants, in percent of GDP)	17.9	14.3	14.4	16.4	15.9	16.9	17.6	18.7	19.1	19.4	20.3	**15.6**	**18.3**
Aid flows (in Million of US dollars) 5/	748.0	799.0	863.9	1003.0	1153.9	1248.9	1186.1	1232.9	1258.4	1609.3	3184.4		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	9.4	10.1	9.8	8.5	7.8	7.2	6.3	5.8	**...**	**7.8**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	75.9	74.2	75.4	75.2	74.4	73.8	73.6	78.4	**...**	**74.3**
Nominal GDP (Million of US dollars)	7,255	7,610	8,136	9,284	9,990	10,941	11,853	13,463	14,725	21,654	48,097		
Nominal dollar GDP growth	-12.0	4.9	6.9	14.1	7.6	9.5	8.3	13.6	9.4	8.1	8.6	**7.0**	**9.3**
Memorandum items:													
PV of external debt 7/	...	...	53.2	49.5	49.4	48.6	47.8	44.6	43.2	38.7	28.7		
In percent of exports	...	...	309.7	296.0	291.1	285.1	279.0	198.1	182.2	150.2	115.3		
Total external debt service-to-exports ratio	3.2	4.0	4.3	10.1	10.5	8.2	9.4	7.0	6.6	6.3	6.3		
PV of PPG external debt (in Million of US dollars)			1879.3	2093.2	2371.0	2687.2	2964.9	3260.0	3567.9	5354.5	9896.5		
(PVt-PVt-1)/GDPt-1 (in percent)				2.6	3.0	3.2	2.5	2.5	2.3	1.9	1.1		
Non-interest current account deficit that stabilizes debt ratio	10.1	12.5	18.0	20.3	21.0	23.3	19.8	17.6	14.9	12.1	11.3		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

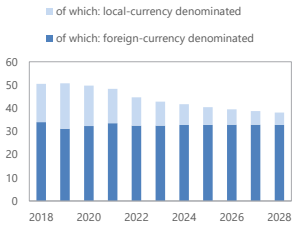
Table 2. Niger: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015-2038

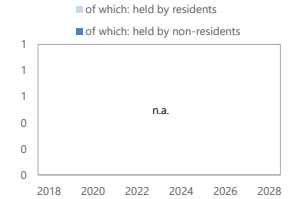
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**39.7**	**43.7**	**49.3**	**50.5**	**50.8**	**49.8**	**48.3**	**44.7**	**42.8**	**38.1**	**34.2**	**29.8**	**44.1**
of which: external debt	**30.1**	**32.6**	**32.4**	**34.0**	**34.9**	**36.1**	**36.6**	**35.4**	**35.2**	**35.2**	**28.5**	**23.5**	**35.4**
Change in public sector debt	9.1	4.0	5.5	1.2	0.3	-1.0	-1.4	-3.6	-1.9	-0.6	-0.4		
Identified debt-creating flows	**10.3**	**4.5**	**-0.2**	**1.0**	**-0.3**	**-0.9**	**-1.0**	**-3.4**	**-1.6**	**-0.7**	**-0.6**	**-0.6**	**-1.0**
Primary deficit	**8.4**	**5.1**	**4.7**	**3.2**	**3.1**	**1.8**	**1.5**	**1.1**	**1.1**	**1.4**	**1.3**	**3.7**	**1.7**
Revenue incl. grants	23.3	20.3	21.2	24.0	24.0	24.9	24.5	24.9	24.9	24.5	25.3	**21.3**	**24.6**
of which: grants	5.4	6.0	6.7	7.6	8.1	8.0	6.9	6.3	5.8	5.0	4.9		
Primary (noninterest) expenditure	31.7	25.4	25.8	27.3	27.1	26.8	26.0	26.1	26.0	25.8	26.6	**25.0**	**26.2**
Automatic debt dynamics	**1.9**	**-0.7**	**-4.7**	**-2.2**	**-3.4**	**-2.7**	**-2.6**	**-4.5**	**-2.8**	**-2.1**	**-1.9**		
Contribution from interest rate/growth differential	-1.3	-1.8	-2.0	-2.5	-2.9	-2.4	-2.3	-4.5	-2.6	-2.1	-1.8		
of which: contribution from average real interest rate	0.0	0.1	0.1	-0.1	0.2	0.4	0.3	0.3	0.2	0.1	0.2		
of which: contribution from real GDP growth	-1.3	-1.9	-2.0	-2.4	-3.1	-2.9	-2.6	-4.8	-2.8	-2.2	-2.0		
Contribution from real exchange rate depreciation	3.2	1.1	-2.8	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**-0.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-3.1**	**0.0**
Privatization receipts (negative)	0.0	0.0	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-1.2**	**-0.5**	**5.7**	**0.5**	**0.1**	**-0.4**	**-0.7**	**-0.3**	**-0.4**	**0.1**	**0.1**	**3.2**	**-0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**38.9**	**39.5**	**39.4**	**38.1**	**36.7**	**33.5**	**31.8**	**27.6**	**26.3**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**183.8**	**164.2**	**164.2**	**152.9**	**149.8**	**134.5**	**127.7**	**112.9**	**103.9**		
Debt service-to-revenue and grants ratio 3/	**5.5**	**11.2**	**19.8**	**28.6**	**33.5**	**37.0**	**35.4**	**30.9**	**27.2**	**14.1**	**15.3**		
Gross financing need 4/	9.4	7.0	8.9	10.1	11.2	11.0	10.2	8.8	7.9	4.8	5.2		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.3	4.9	4.9	5.2	6.5	6.0	5.6	11.0	6.8	6.0	6.1	**5.8**	**6.4**
Average nominal interest rate on external debt (in percent)	0.0	0.0	0.0	1.5	1.5	1.5	1.5	1.5	1.5	1.6	1.7	**1.4**	**1.6**
Average real interest rate on domestic debt (in percent)	4.3	4.0	6.2	1.0	2.5	3.1	2.8	3.4	3.0	3.6	4.1	**1.9**	**3.0**
Real exchange rate depreciation (in percent, + indicates depreciation)	13.4	4.0	-9.1	...	...	...	...	...	...	...	...	**1.2**	**...**
Inflation rate (GDP deflator, in percent)	1.0	0.2	-0.1	3.3	2.4	1.9	2.0	1.4	2.0	2.0	2.4	**2.8**	**2.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	8.0	-16.0	6.6	11.1	6.0	4.6	2.6	11.2	6.6	5.8	6.4	**7.8**	**6.5**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-0.7	1.1	-0.9	2.0	2.9	2.8	2.9	4.8	3.0	2.0	1.7	**-0.2**	**2.7**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
				16.4	15.9	16.9	17.6	18.7	19.1	19.4	20.3		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The general government, central bank, government-guaranteed debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

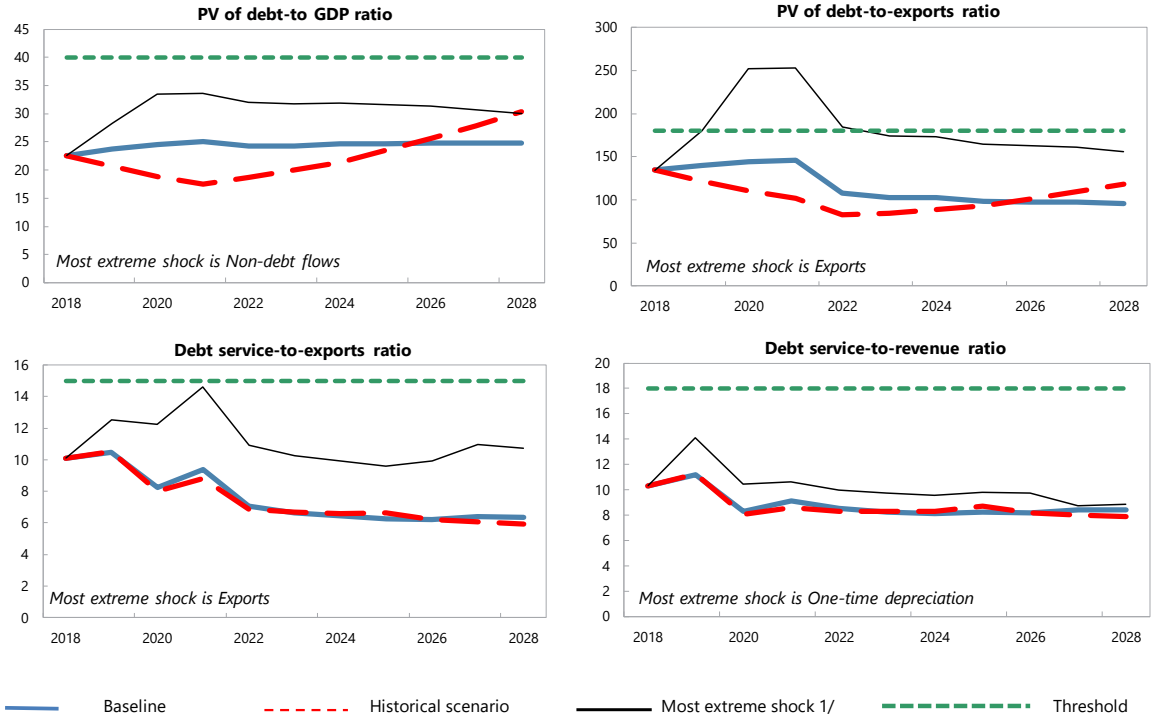
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Niger: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2018-2028



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.8%	1.8%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	6	6

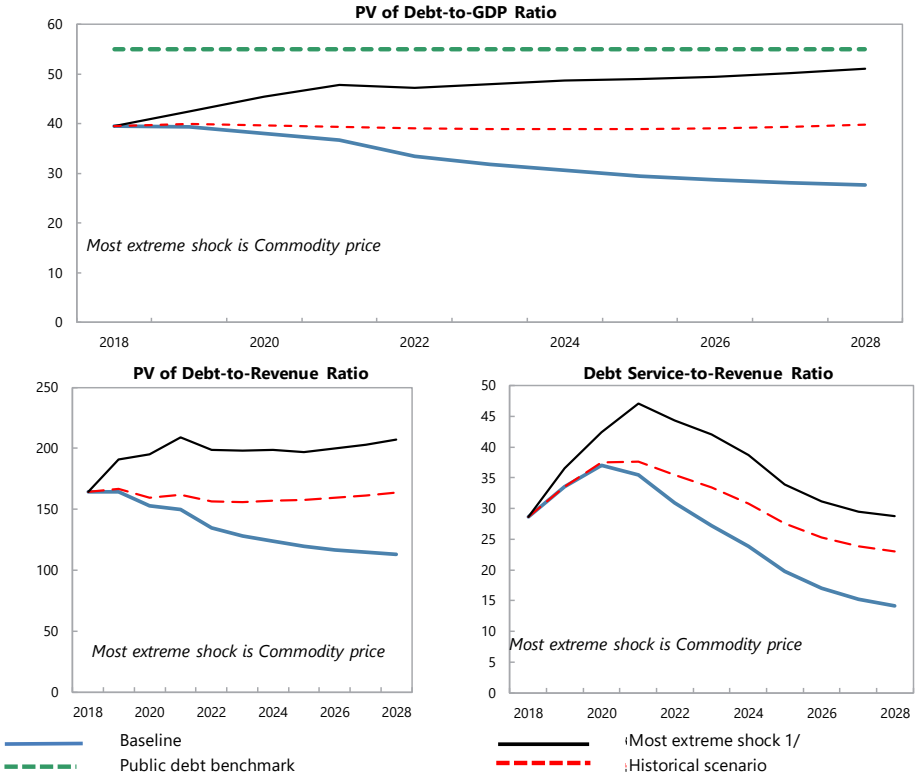
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Niger: Indicators of Public Debt Under Alternative Scenarios, 2018-2028



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	55%	55%
Domestic medium and long-term	17%	17%
Domestic short-term	27%	27%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.8%	1.8%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.6%	4.2%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	3%	3.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Niger: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018-2028

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	23	24	25	25	24	24	25	25	25	25	25
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	23	21	19	17	19	20	21	23	26	28	30
B. Bound Tests											
B1. Real GDP growth	23	25	28	28	28	28	28	28	28	28	28
B2. Primary balance	23	25	28	29	28	28	29	29	29	29	28
B3. Exports	23	26	30	31	29	29	29	29	29	29	28
B4. Other flows 3/	23	28	33	34	32	32	32	32	31	31	30
B5. One-time 30 percent nominal depreciation	23	30	25	26	25	25	26	26	27	27	27
B6. Combination of B1-B5	23	30	32	32	31	31	31	31	31	31	30
C. Tailored Tests											
C1. Combined contingent liabilities	23	26	28	28	28	28	28	28	28	28	28
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	23	26	29	29	28	27	27	27	27	26	25
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	135	140	144	146	108	102	103	98	98	98	96
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	135	122	111	102	83	84	89	93	101	110	118
B. Bound Tests											
B1. Real GDP growth	135	140	144	146	108	102	103	98	98	98	96
B2. Primary balance	135	148	165	170	125	120	120	115	114	113	110
B3. Exports	135	**180**	**252**	**253**	**185**	174	173	165	163	161	156
B4. Other flows 3/	135	167	**196**	**196**	142	134	133	126	123	121	117
B5. One-time 30 percent nominal depreciation	135	140	115	118	88	85	86	83	83	84	84
B6. Combination of B1-B5	135	178	167	**204**	149	141	141	134	132	131	127
C. Tailored Tests											
C1. Combined contingent liabilities	135	155	163	165	123	117	117	111	111	110	108
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	135	**182**	**195**	**189**	131	120	116	109	106	104	101
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	10	10	8	9	7	7	6	6	6	6	6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	10	11	8	9	7	7	7	7	6	6	6
B. Bound Tests											
B1. Real GDP growth	10	10	8	9	7	7	6	6	6	6	6
B2. Primary balance	10	10	8	10	7	7	7	7	7	7	7
B3. Exports	10	13	12	15	11	10	10	10	10	11	11
B4. Other flows 3/	10	10	9	11	8	7	7	7	8	8	8
B5. One-time 30 percent nominal depreciation	10	10	8	9	7	6	6	6	6	5	5
B6. Combination of B1-B5	10	12	11	12	9	8	8	8	9	9	9
C. Tailored Tests											
C1. Combined contingent liabilities	10	10	9	10	7	7	7	7	7	7	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	10	13	10	11	8	7	7	7	7	8	7
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	10	11	8	9	9	8	8	8	8	8	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2028 2/	10	11	8	9	8	8	8	9	8	8	8
B. Bound Tests											
B1. Real GDP growth	10	12	9	10	10	9	9	9	9	10	10
B2. Primary balance	10	11	8	10	9	9	9	9	9	10	10
B3. Exports	10	11	9	10	9	9	9	9	9	10	10
B4. Other flows 3/	10	11	9	10	10	9	9	9	10	11	11
B5. One-time 30 percent nominal depreciation	10	14	10	11	10	10	10	10	10	9	9
B6. Combination of B1-B5	10	12	10	11	10	10	9	10	10	11	11
C. Tailored Tests											
C1. Combined contingent liabilities	10	11	9	10	9	9	9	9	9	9	9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	10	13	10	11	10	9	9	9	9	10	9
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Niger: Sensitivity Analysis for Key Indicators of Public Debt, 2018-2028

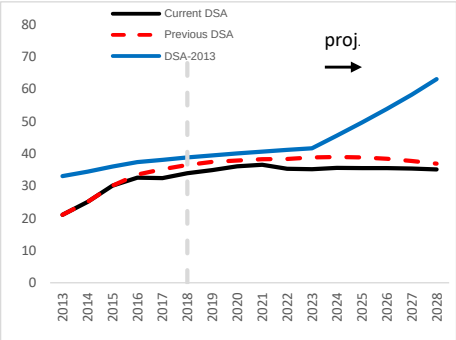
	Projections										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	39	39	38	37	34	32	31	30	29	28	28
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	39	40	40	39	39	39	39	39	39	39	40
B. Bound Tests											
B1. Real GDP growth	39	43	46	46	44	43	43	43	44	45	45
B2. Primary balance	39	42	45	43	39	36	35	34	32	32	31
B3. Exports	39	41	43	42	38	36	35	34	33	32	31
B4. Other flows 2/	39	44	47	45	41	39	38	36	35	34	33
B6. One-time 30 percent nominal depreciation	39	44	40	37	32	29	27	25	23	21	19
B6. Combination of B1-B5	39	41	41	38	34	32	31	29	28	28	27
C. Tailored Tests											
C1. Combined contingent liabilities	39	45	43	41	37	35	34	33	32	31	30
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	39	42	44	46	45	46	47	47	48	49	50
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of Debt-to-Revenue Ratio											
Baseline	164	164	153	150	134	128	124	120	117	115	113
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	164	166	160	162	156	156	157	158	159	161	163
B. Bound Tests											
B1. Real GDP growth	164	175	176	180	169	168	170	171	174	177	180
B2. Primary balance	164	177	179	174	155	146	141	136	132	129	126
B3. Exports	164	172	174	170	153	145	141	136	132	129	126
B4. Other flows 2/	164	183	188	185	166	158	153	148	143	139	135
B6. One-time 30 percent nominal depreciation	164	187	166	156	133	120	111	101	93	86	80
B6. Combination of B1-B5	164	171	166	155	137	129	123	118	115	112	110
C. Tailored Tests											
C1. Combined contingent liabilities	164	188	173	168	150	142	137	132	128	125	123
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	164	184	186	198	188	188	190	191	194	198	203
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	29	34	37	35	31	27	24	20	17	15	14
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 1/	29	34	38	38	36	33	31	28	25	24	23
B. Bound Tests											
B1. Real GDP growth	29	35	42	42	39	36	33	30	27	26	25
B2. Primary balance	29	34	41	44	37	32	26	22	19	17	16
B3. Exports	29	34	37	36	31	28	24	20	18	16	15
B4. Other flows 2/	29	34	37	36	32	28	24	20	18	17	16
B6. One-time 30 percent nominal depreciation	29	33	37	35	31	28	25	21	18	16	15
B6. Combination of B1-B5	29	33	37	39	33	29	25	20	17	15	14
C. Tailored Tests											
C1. Combined contingent liabilities	29	34	45	40	35	30	25	21	18	16	15
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	29	36	42	45	43	40	37	33	30	29	28
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
2/ Includes official and private transfers and FDI.

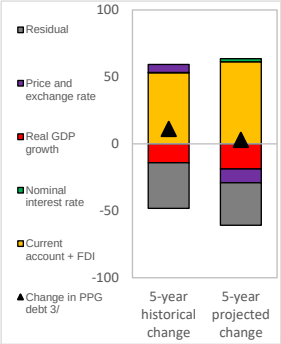
Figure 3. Niger: Drivers of Debt Dynamics – Baseline Scenario External Debt

External Debt

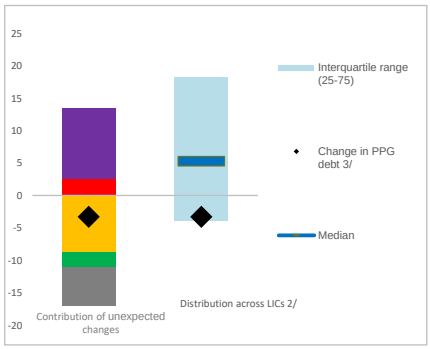
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

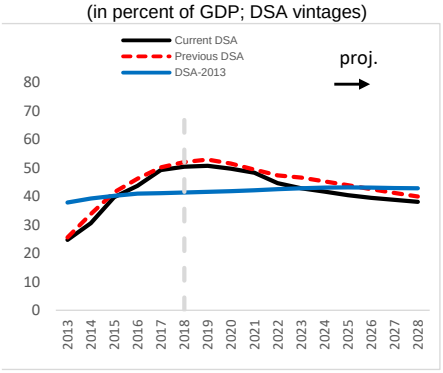


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

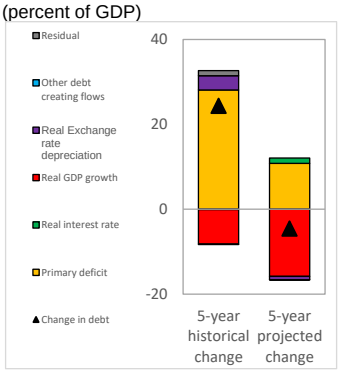


Public Debt

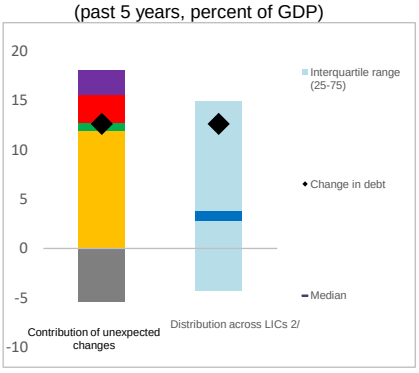
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



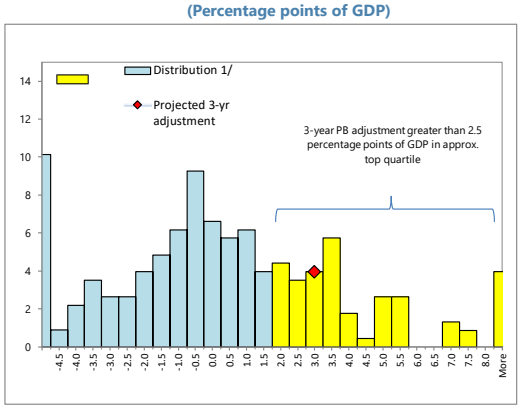
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

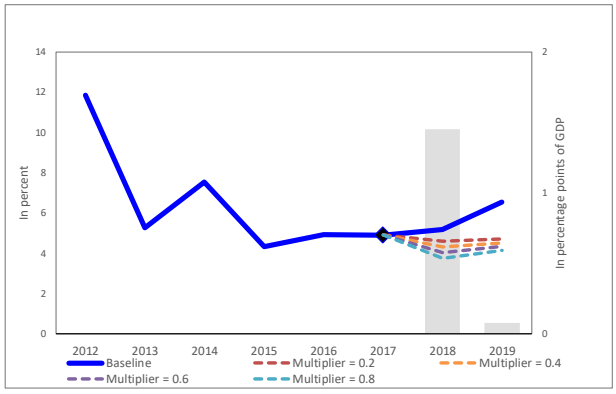
Figure 4. Niger: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

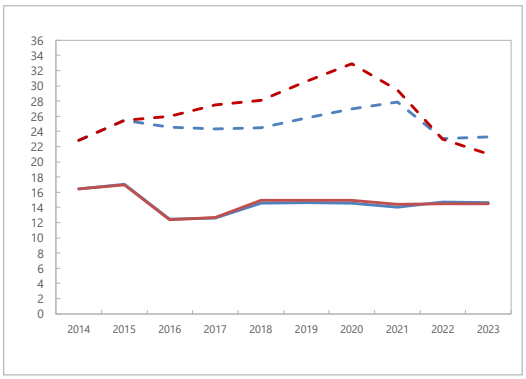
Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates

(% of GDP)



Contribution to Real GDP growth

(percent, 5-year average)

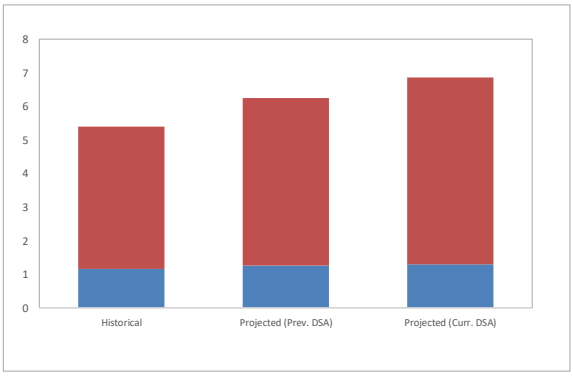


Figure 5. Niger: Qualification of the Moderate Category, 2018-2028 1/

Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.